Negative Binomial models, intro to ZIP models

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Project

Assignment: conduct an analysis with the provided dataset to answer the research question of interest.

- Can be completed individually or in groups
  - More expectations for groups (more models and more deliverables)
  - Groups will be randomly assigned
  - Tell me by **October 27** whether you want to be in a group
- Completed in several stages

Project timeline

- **October 27:** Email me to tell me whether you will complete the project individually, or wish to work in a group. I will randomly create groups of 2-3 students from all those who elect to work in a group.
- **November 6:** Statistical analysis plan (15% of grade)
- **November 13:** Exploratory data analysis (15% of grade)
- **November 16-20:** Check-in appointment with Dr. Evans (15% of grade)
- **December 4:** Final submission (55% of grade)

Final submission

For groups:

- an extended written report (in the style of a journal article)
- a 10-15 minute presentation to Dr. Evans, plus questions

For individuals: choose *one* of the following

- an extended written report (in the style of a journal article)
- OR a short written report (magazine article) + a 10-15 minute presentation to Dr. Evans (plus questions)

Overdispersion

Overdispersion occurs when the response variable Y_i has greater variability than the model accounts for.

Poisson: $Var(Y_i|\mathbf{x}_i) = \mathbb{E}[Y_i|\mathbf{x}_i] = \mu_i$

Negative Binomial models

$(\phi = 1)$

known constant

$$Y \sim NB(\mu, r) \qquad f(y;\mu,r) = \frac{\Gamma(y+r)}{\Gamma(y+1)\Gamma(r)}\left(\frac{\mu}{\mu+r}\right)^y\left(1-\frac{\mu}{\mu+r}\right)^r$$

$(r > 0)$

$$Y_i|\mathbf{x}_i \sim NB(\mu_i, r) \qquad \log(\mu_i) = \mathbf{x}_i^T\beta$$

$$\mathrm{var}(Y_i|x_i) = \mu_i + \frac{\mu_i^2}{r}$$

For large r, $\mathrm{var}(Y_i|x_i) \approx \mu_i$

For small r, $\mathrm{var}(Y_i|x_i) >> \mu_i$

In practice, r is unknown

$\Rightarrow$ not an EDM, have to estimate r

Example: articles data

specifically for NB response

don't specify a family

```
m3 <- MASS::glm.nb(art ~ phd + ment + kid5, data = articles
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.233528   0.117585   1.986   0.0470 *
phd          0.008203   0.036069   0.227   0.8201
ment         0.030374   0.003240   9.373   <2e-16 ***
kid5        -0.093256   0.046475  -2.007   0.0448 *

(Dispersion parameter for Negative Binomial(2.1647)
   family taken to be 1)

AIC: 3145.6
```

$\hat{r}$

$\phi = 1$

Inference for negative binomial models

- Can use the usual Wald tests for β (from asymptotic normality)
- LRT also works because a full distribution is specified

LRT: $-2\log\Lambda = 2\log L_{full} - 2\log L_{reduced} \approx \chi^2_{p_{full}-p_{reduced}}$

If r is known, then we could compare deviances

But if r is unknown, then $\hat{r}_{full}$ & $\hat{r}_{reduced}$ will be different $\Rightarrow$ can't just compare deviances

EDM: $f(y;\mu,\phi) = b(y,\phi)\exp\left\{-\frac{d(y,\mu)}{2\phi}\right\}$

For NB, this depends on r

$\Rightarrow$ $b(y,\phi)$ will be different for full & reduced models

Example: articles data

```r
m3 <- MASS::glm.nb(art ~ phd + ment + kid5, data = articles
m4 <- MASS::glm.nb(art ~ phd + kid5, data = articles)

pchisq(m3$twologlik - m4$twologlik, 1, lower.tail=F)
```

```
## [1] 1.708206e-18
```

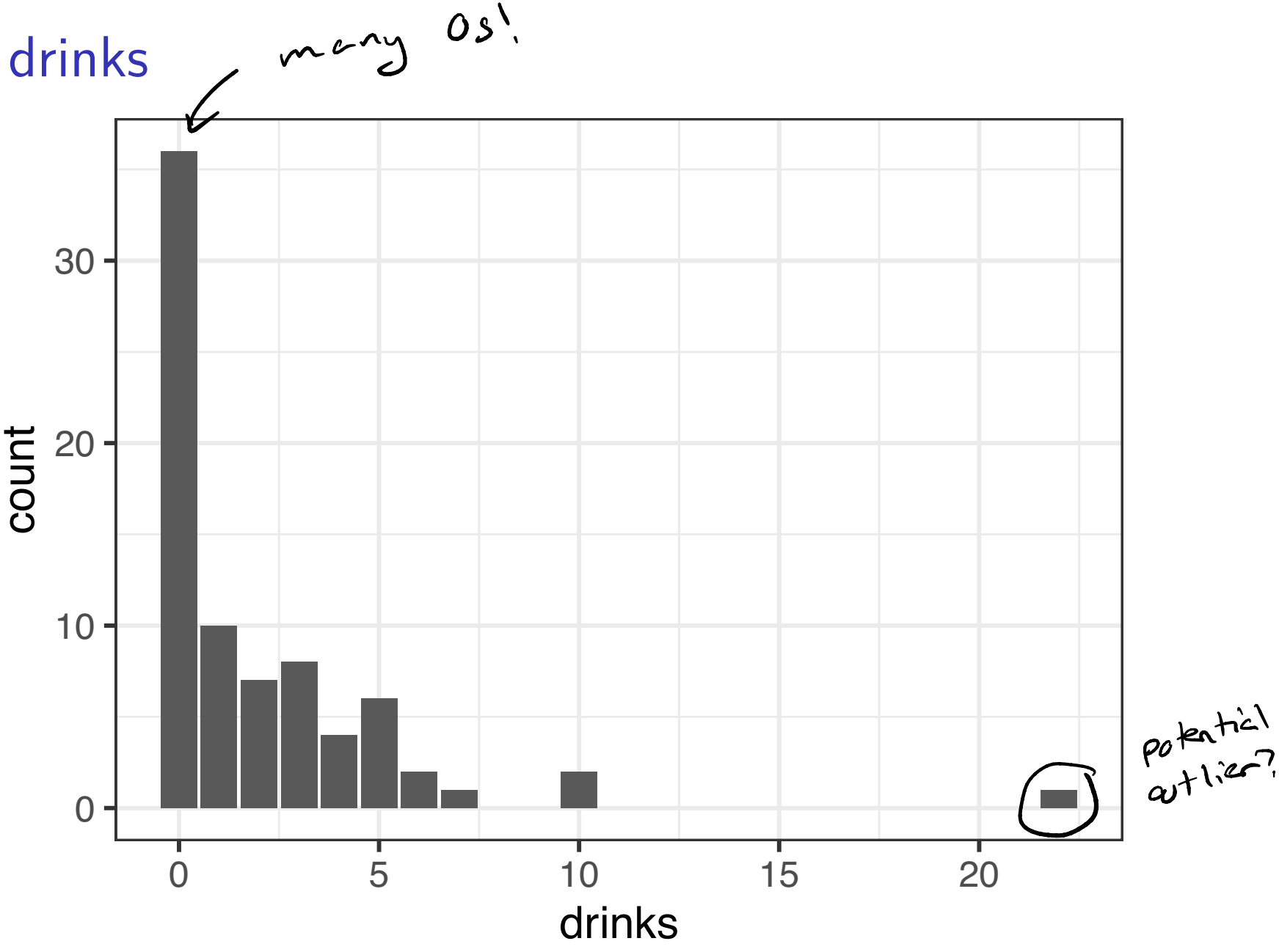
New data

Survey data from 77 college students on a dry campus (i.e., alcohol is prohibited) in the US. Survey asks students "How many alcoholic drinks did you consume last weekend?"

- `drinks`: number of drinks the student reports consuming
- `sex`: whether the student identifies as male
- `OffCampus`: whether the student lives off campus
- `FirstYear`: whether the student is a first-year student

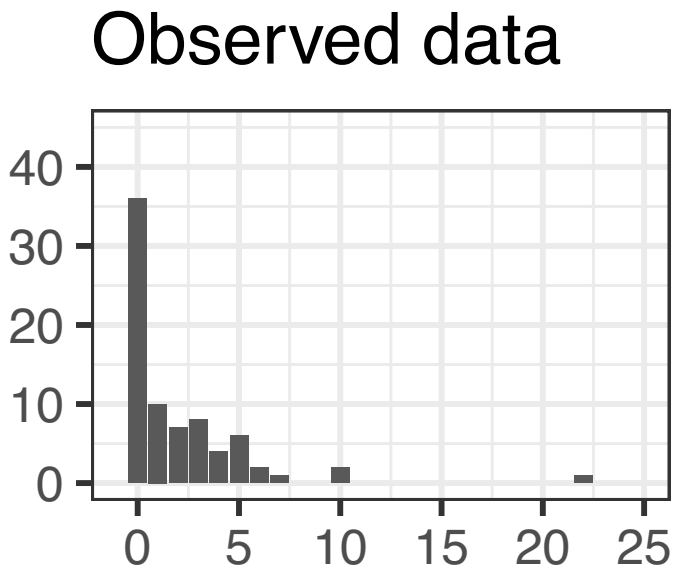
Our goal: model the number of drinks students report consuming.

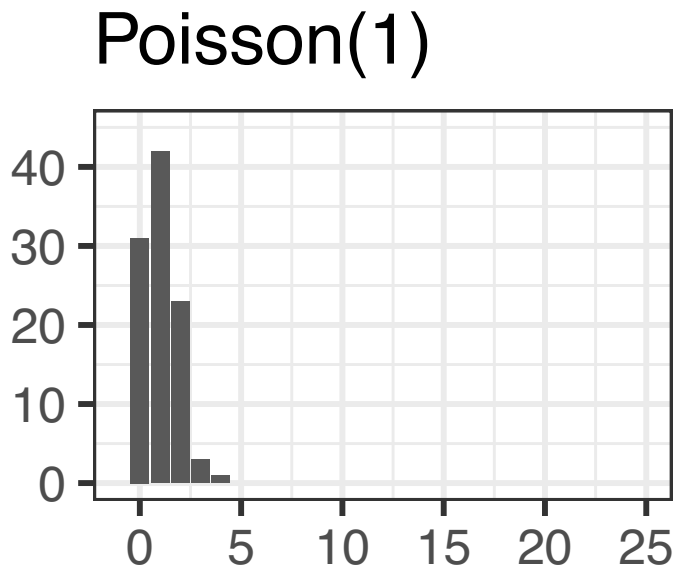
EDA: drinks

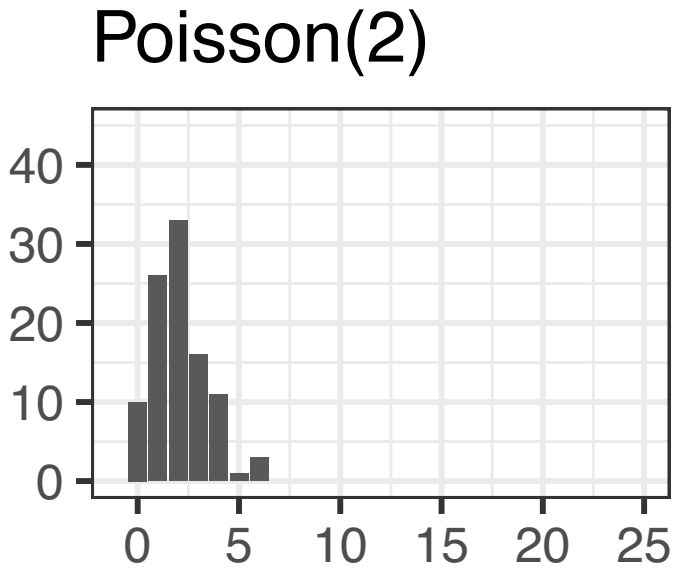


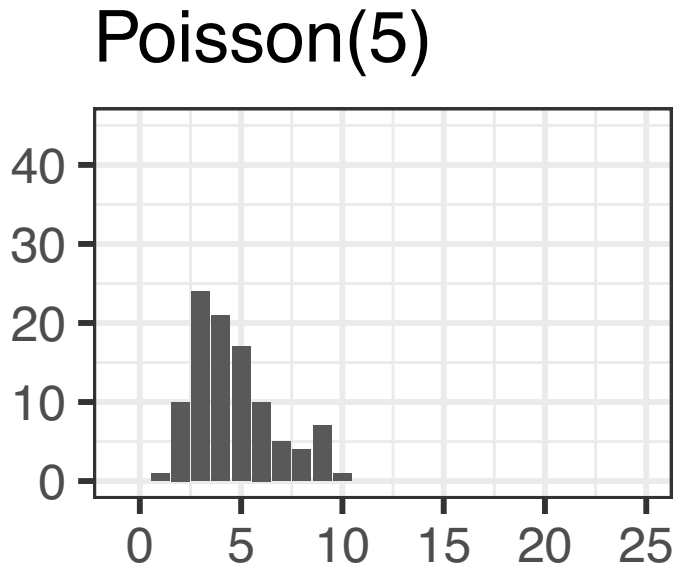
What do you notice about this distribution?

Comparisons with Poisson distributions







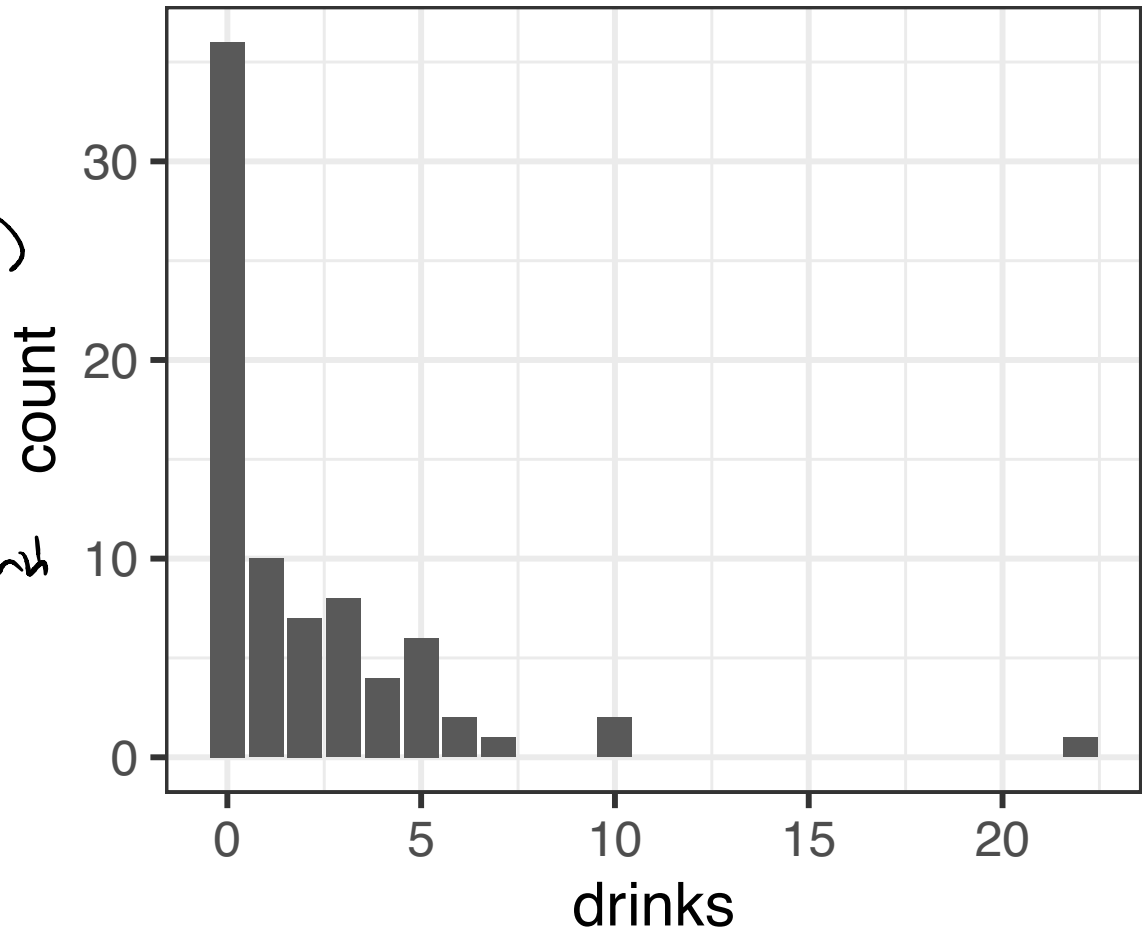


Excess zeros

one option: Students are not being truthful (even though anonymous survey)

Why might there be excess 0s in the data, and why is that a problem for modeling the number of drinks consumed?

Two groups:
- students who never drink (always 0s) "structural zero"
- students who sometimes drink



Zero-inflated Poisson (ZIP) model

Y_i = # drinks consumed by student i

Two possibilities:

- Student i never drinks
- Student i sometimes drinks

$Z_i = 1$ } unobserved (latent variable)
$Z_i = 0$ }

$$Y_i \mid Z_i = 0, x_i \sim \text{Poisson}(\lambda_i)$$

$$Y_i \mid Z_i = 1, x_i = 0 \quad \text{(point mass at 0)}$$

$$\log(\lambda_i) = x_i^T \beta$$

$$p_i = P(Z_i = 1 \mid x_i)$$

$$\log\left(\frac{p_i}{1-p_i}\right) = x_i^T \gamma$$

Zero-inflated Poisson (ZIP) model

$$P(Y_i = y \mid Z_i = 1, x_i) = \begin{cases} 1 & y = 0 \\ 0 & y > 0 \end{cases}$$

$$P(Y_i = y \mid Z_i = 0, x_i) = \frac{e^{-\lambda_i} \lambda_i^{y}}{y!}$$

$1 - p_i$

$$P(Y_i = y \mid x_i) = P(Y_i = y \mid Z_i = 0, x_i)\, P(Z_i = 0 \mid x_i) + P(Y_i = y \mid Z_i = 1, x_i)\, \underbrace{P(Z_i = 1 \mid x_i)}_{p_i}$$

$$= \begin{cases} p_i + (1 - p_i) \underbrace{e^{-\lambda_i}}_{P(Y_i = 0 \mid Z_i = 0, x_i)} & y = 0 \\ (1 - p_i) \underbrace{\dfrac{e^{-\lambda_i} \lambda_i^{y}}{y!}}_{P(Y_i = y \mid Z_i = 0, x_i)} & y > 0 \end{cases}$$

In R

```
library(pscl)

m1 <- zeroinfl(drinks ~ sex + FirstYear + OffCampus,
               dist = "poisson", data = wdrinks)

m1$coefficients
```

$\log(\hat{\lambda}_i)$

```
## $count
##   (Intercept)           sexm FirstYearTRUE OffCampusTRUE
##     0.8010435      0.9834693    -0.1619318     0.3723520
##
## $zero
##   (Intercept)           sexm FirstYearTRUE OffCampusTRUE
##   -0.39618515   -0.07079078     0.89197114   -1.69137146
```

$\log\left(\frac{\hat{p}_i}{1-\hat{p}_i}\right)$

Activity

Work on the activity (handout).